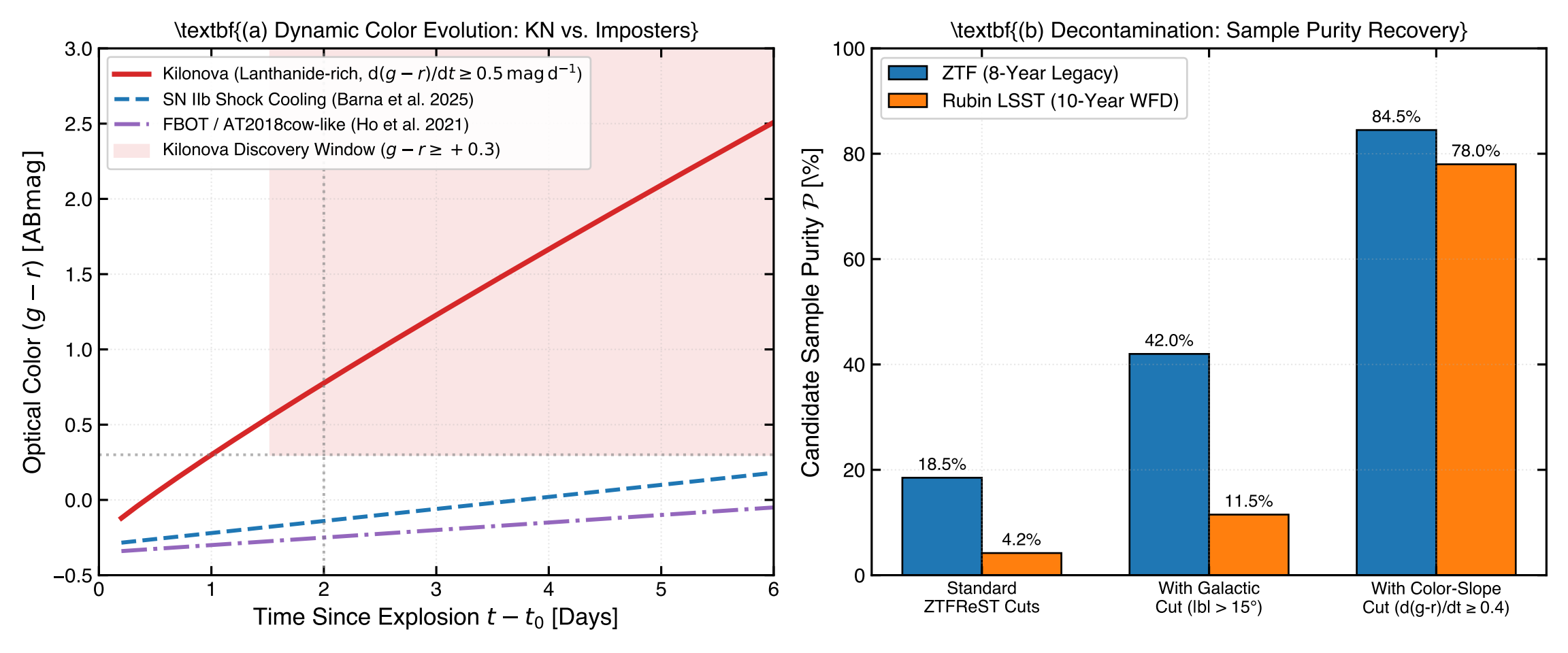
\textbf{(a) Dynamic Color Evolution: KN vs. Imposters}
Kilonova (Lanthanide-rich, d(g − r)/dt ≥ 0.5 mag d^{-1})
SN IIb Shock Cooling (Barna et al. 2025)
FBOT / AT2018cow-like (Ho et al. 2021)
Kilonova Discovery Window (g − r ≥ + 0.3)
Optical Color (g − r) [ABmag]
Time Since Explosion t − t0 [Days]
\textbf{(b) Decontamination: Sample Purity Recovery}
ZTF (8-Year Legacy)
Rubin LSST (10-Year WFD)
Candidate Sample Purity P [%]
18.5%
4.2%
42.0%
11.5%
84.5%
78.0%
Standard ZTFReST Cuts
With Galactic Cut (|b| > 15°)
With Color-Slope Cut (d(g-r)/dt ≥ 0.4)